

ASSIGNMENT NO.8: Views and Data Abstraction

Questions:

Q1) Create a MySQL database named CollegeDB and create two tables: Course and Student.

Course Table

Column Name	Data Type	Constraint
CourseID	INT	PRIMARY KEY
CourseName	VARCHAR(50)	NOT NULL
Fees	DECIMAL(10,2)	NOT NULL

Student Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
CourseID	INT	FOREIGN KEY
City	VARCHAR(30)	
Marks	INT	
Email	VARCHAR(60)	

Insert at least 4 course records and 10 student records.

Answer:-

MySQL query

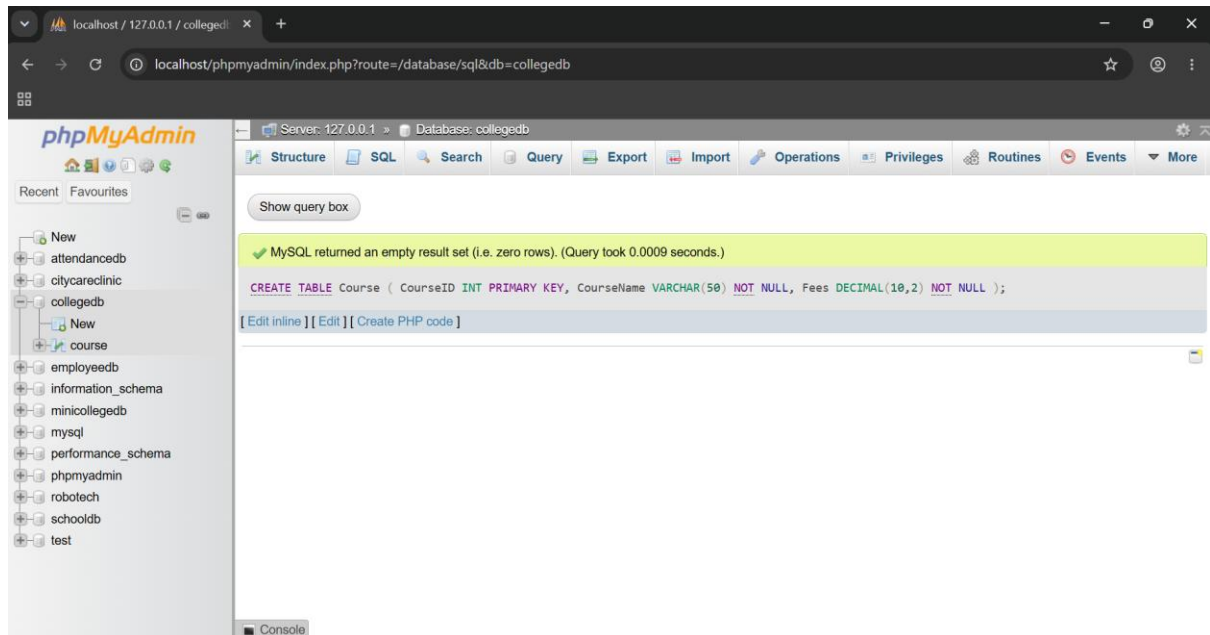
```
CREATE TABLE Course (
```

```
    CourseID INT PRIMARY KEY,
```

```
    CourseName VARCHAR(50) NOT NULL,
```

Fees DECIMAL(10,2) NOT NULL

);



CREATE TABLE Student (

StudentID INT PRIMARY KEY,

StudentName VARCHAR(50) NOT NULL,

CourseID INT,

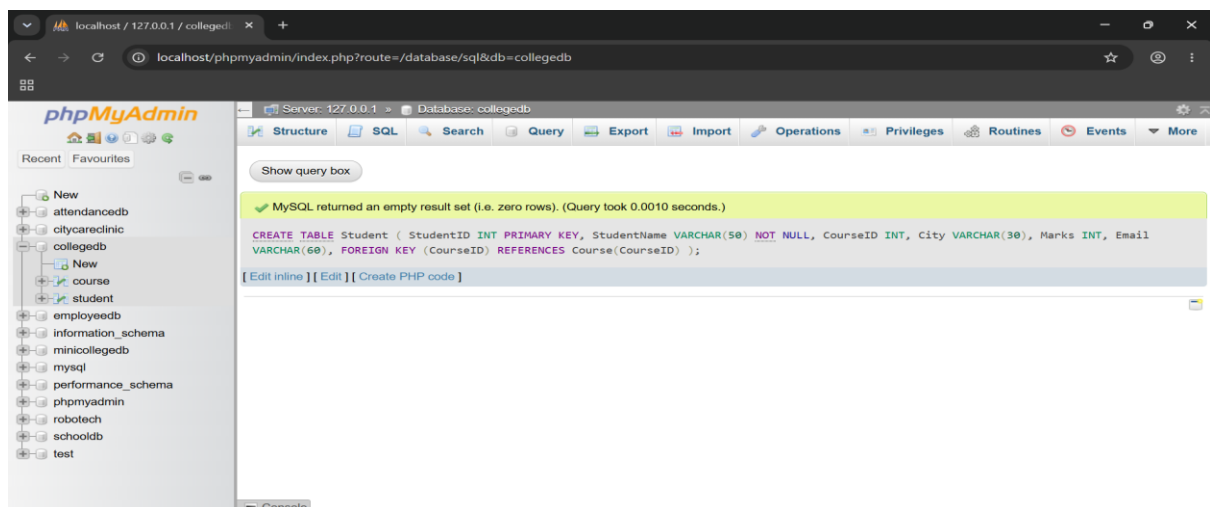
City VARCHAR(30),

Marks INT,

Email VARCHAR(60),

FOREIGN KEY (CourseID) REFERENCES Course(CourseID)

);



localhost / 127.0.0.1 / collegedi x +

localhost/phpmyadmin/index.php?route=/database/sql&db=collegedb

phpMyAdmin

Recent Favourites

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- attendancedb
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- employeedb
- information_schema
- minicollegedb
- mysql
- performance_schema
- phpmyadmin
- robotech
- schooldb
- test

Server: 127.0.0.1 > Database: collegedb

Structure SQL Search Query Export Import Operations Privileges Routines Events More

Show query box

4 rows inserted. (Query took 0.0054 seconds.)

```
INSERT INTO Course (CourseID, CourseName, Fees) VALUES (101, 'BSc IT', 45000.00), (102, 'BCA', 40000.00), (103, 'BSc CS', 42000.00), (104, 'Data Science', 55000.00);
```

[Edit inline] [Edit] [Create PHP code]

localhost/phpmyadmin/index.php?route=/server/privileges&db=collegedb&

localhost / 127.0.0.1 / collegedi x +

localhost/phpmyadmin/index.php?route=/database/sql&db=collegedb

phpMyAdmin

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- test

Server: 127.0.0.1 > Database: collegedb

Structure SQL Search Query Export Import Operations Privileges Routines Events More

Show query box

10 rows inserted. (Query took 0.0011 seconds.)

```
INSERT INTO Student (StudentID, StudentName, CourseID, City, Marks, Email) VALUES (1, 'Rahul Sharma', 101, 'Mumbai', 78, 'rahul@gmail.com'), (2, 'Priya Patil', 102, 'Pune', 85, 'priya@gmail.com'), (3, 'Aman Verma', 103, 'Nashik', 67, 'aman@gmail.com'), (4, 'Sneha Joshi', 101, 'Mumbai', 91, 'sneha@gmail.com'), (5, 'Karan Mehta', 102, 'Thane', 74, 'karan@gmail.com'), (6, 'Neha Shah', 101, 'Pune', 88, 'neha@gmail.com'), (7, 'Rohan Desai', 103, 'Mumbai', 69, 'rohan@gmail.com'), (8, 'Meera Kulkarni', 102, 'Nashik', 82, 'meera@gmail.com'), (9, 'Arjun Nair', 101, 'Thane', 59, 'arjun@gmail.com'), (10, 'Divya More', 102, 'Mumbai', 93, 'divya@gmail.com');
```

[Edit inline] [Edit] [Create PHP code]

Console

localhost / 127.0.0.1 / collegedi x +

localhost/phpmyadmin/index.php?route=/database/sql&db=collegedb

phpMyAdmin

Recent Favourites

- New
- attendancedb
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 - student
- employeedb
- information_schema
- minicollegedb
- mysql
- performance_schema
- phpmyadmin
- robotech
- schooldb
- test

Server: 127.0.0.1 > Database: collegedb > Table: Course

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More

Show query box

Showing rows 0 - 3 (4 total, Query took 0.0014 seconds.)

```
SELECT * FROM Course;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	CourseID	CourseName	Fees
☐ Edit ☐ Copy ☐ Delete	101	BSc IT	45000.00
☐ Edit ☐ Copy ☐ Delete	102	BCA	40000.00
☐ Edit ☐ Copy ☐ Delete	103	BSc CS	42000.00
☐ Edit ☐ Copy ☐ Delete	104	Data Science	55000.00

☐ Check all | With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

localhost/phpmyadmin/index.php?route=/server/privileges&db=collegedb&table=Course&checkprivsdb=collegedb&checkprivtable=Course&viewing_mode=table None

Server: 127.0.0.1 » Database: collegedb » Table: Student

Showing rows 1 - 10 (10 total, Query took 0.0010 seconds.)

```
SELECT * FROM Student;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	StudentID	StudentName	CourseID	City	Marks	Email
☐ Edit ☐ Copy ☐ Delete	1	Rahul Sharma	101	Mumbai	78	rahul@gmail.com
☐ Edit ☐ Copy ☐ Delete	2	Priya Patil	102	Pune	85	priya@gmail.com
☐ Edit ☐ Copy ☐ Delete	3	Aman Verma	103	Nashik	67	aman@gmail.com
☐ Edit ☐ Copy ☐ Delete	4	Sneha Joshi	101	Mumbai	91	sneha@gmail.com
☐ Edit ☐ Copy ☐ Delete	5	Karan Mehta	102	Thane	74	karan@gmail.com
☐ Edit ☐ Copy ☐ Delete	6	Neha Shah	101	Pune	88	neha@gmail.com
☐ Edit ☐ Copy ☐ Delete	7	Rohan Desai	103	Mumbai	69	rohan@gmail.com
☐ Edit ☐ Copy ☐ Delete	8	Meera Kulkarni	102	Nashik	82	meera@gmail.com
☐ Edit ☐ Copy ☐ Delete	9	Arjun Nair	101	Thane	59	arjun@gmail.com
☐ Edit ☐ Copy ☐ Delete	10	Divya More	102	Mumbai	93	divya@gmail.com

Q2) Create a view named StudentBasicView to display only selected student details.

The view should contain:

Column
StudentID
StudentName
Column
City
Marks

Answer:-

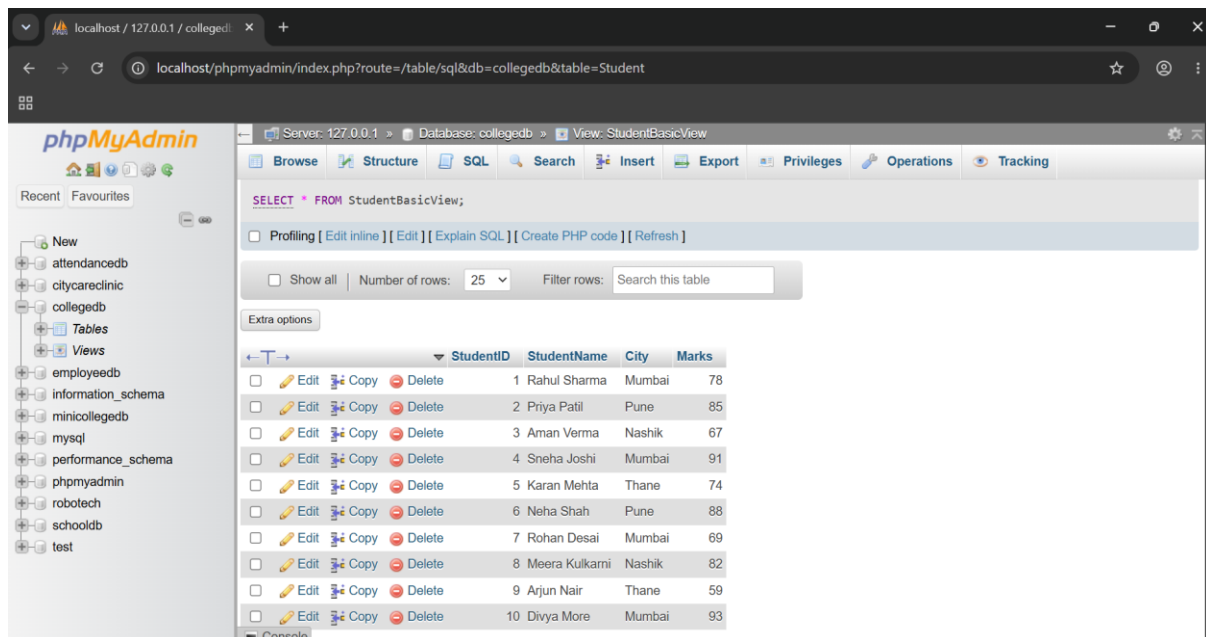
Server: 127.0.0.1 » Database: collegedb » Table: Student

Show query box

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0013 seconds.)

```
CREATE VIEW StudentBasicView AS SELECT StudentID, StudentName, City, Marks FROM Student;
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

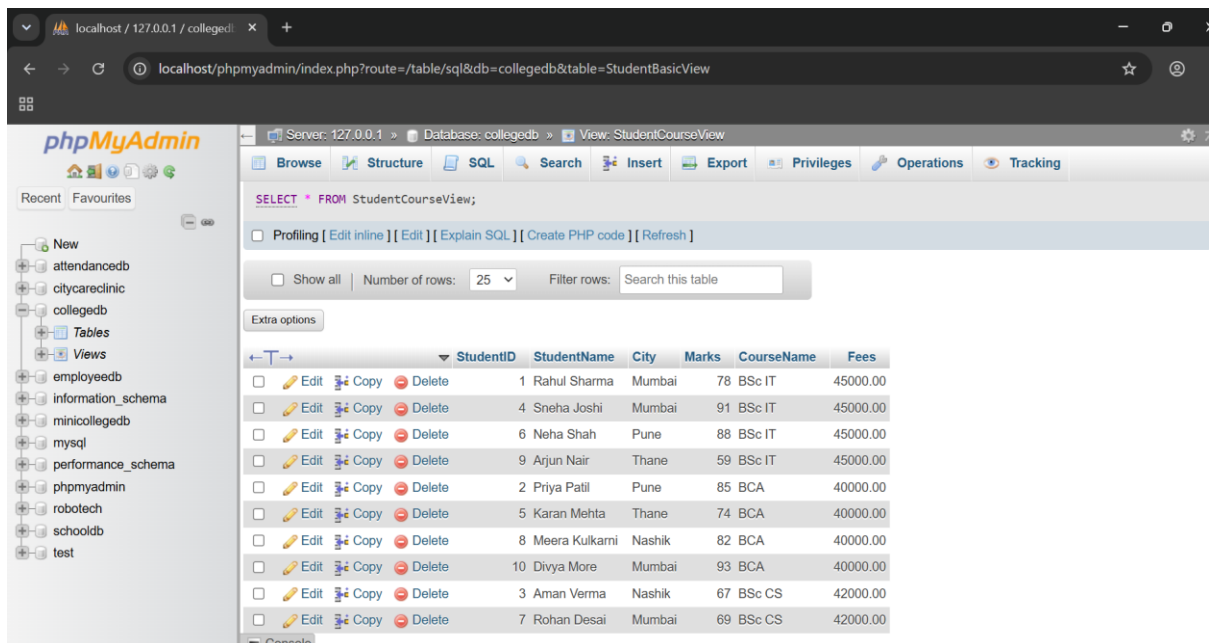
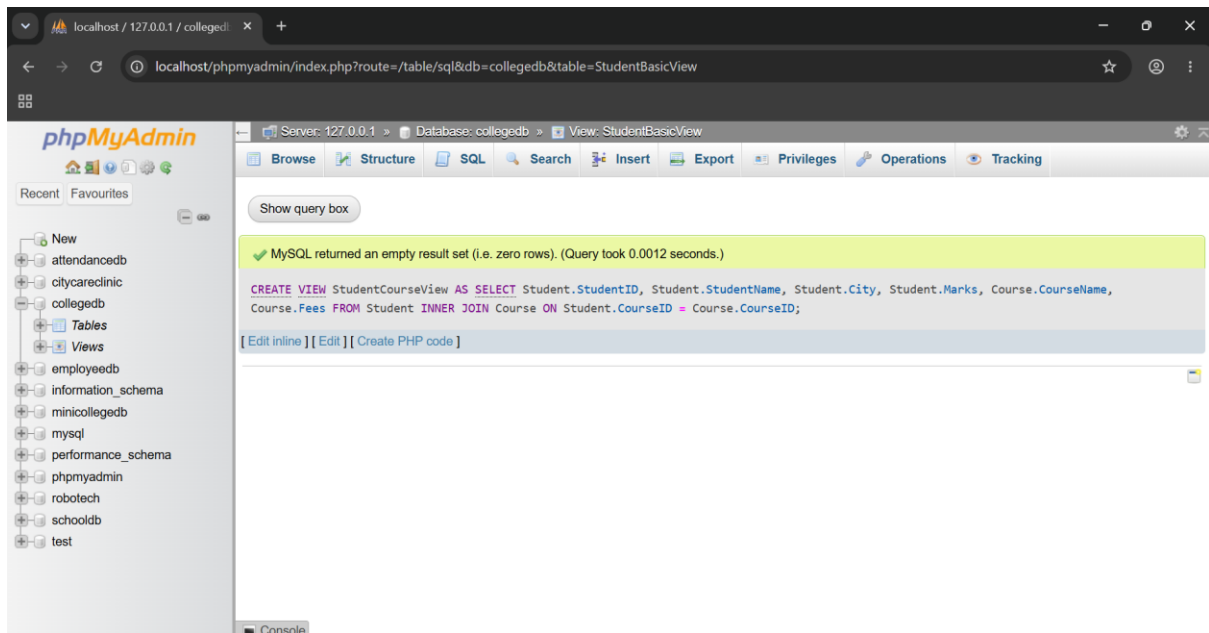


Q3) Create a view named StudentCourseView using Student and Course tables.

The view should display:

Column
StudentID
StudentName
City
Marks
CourseName
Fees

Answer:-



Q4) Write a MySQL query using StudentCourseView to display students from Mumbai. The output should contain:

Column
StudentName
City
CourseName

Marks

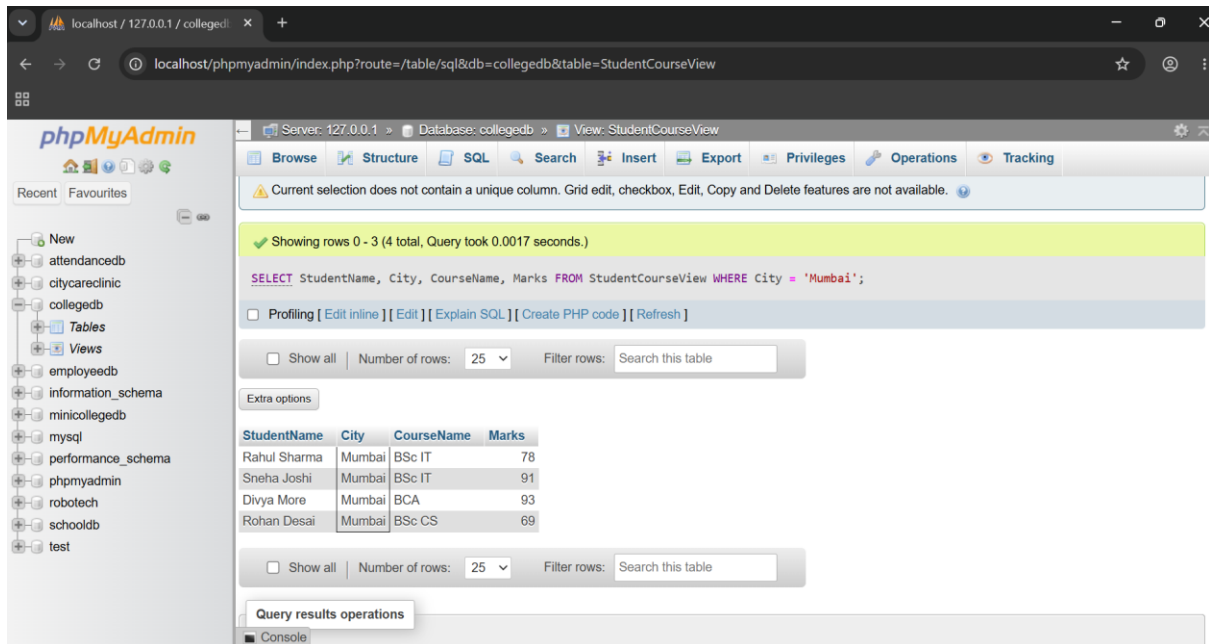
Answer:-

MySQL query

SELECT StudentName, City, CourseName, Marks

FROM StudentCourseView

WHERE City = 'Mumbai';



Q5) Write a MySQL query using StudentCourseView to display students sorted by marks in descending order.

The output should contain:

Column

StudentName

CourseName

Marks

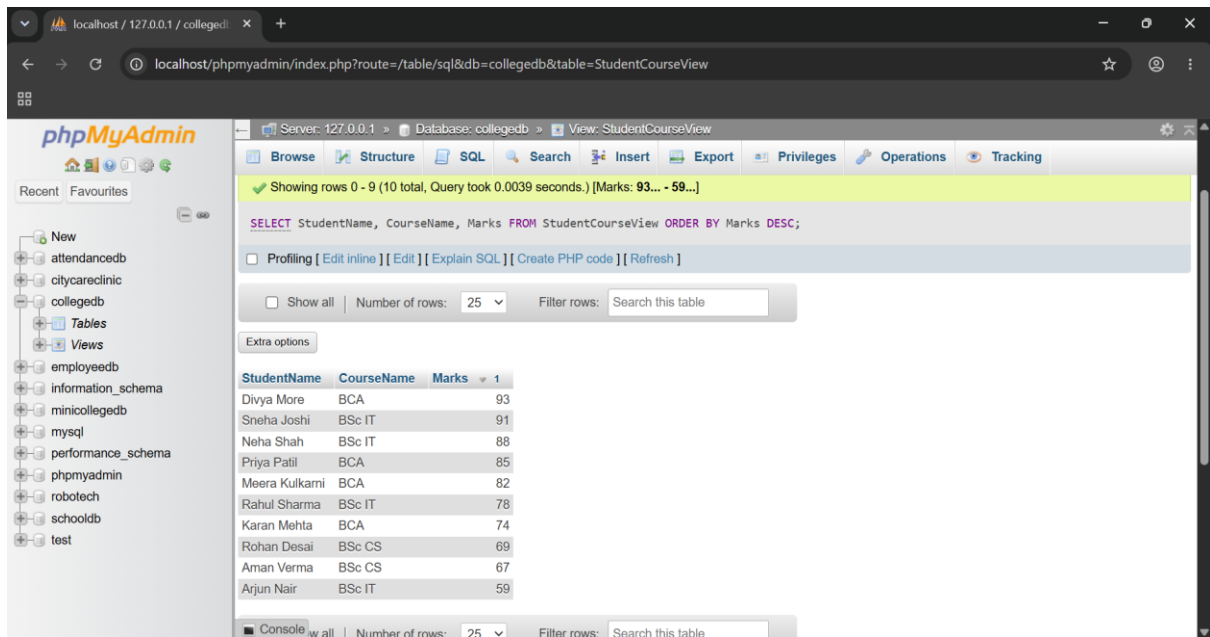
Answer:-

MySQL query

SELECT StudentName, CourseName, Marks

FROM StudentCourseView

ORDER BY Marks DESC;



The screenshot shows the phpMyAdmin interface for a database named 'collegedb'. The 'StudentCourseView' table is selected, and the SQL query 'SELECT StudentName, CourseName, Marks FROM StudentCourseView ORDER BY Marks DESC;' is executed. The results show 10 rows, sorted by Marks in descending order. The table has columns: StudentName, CourseName, and Marks.

StudentName	CourseName	Marks
Divya More	BCA	93
Sneha Joshi	BSc IT	91
Neha Shah	BSc IT	88
Priya Patil	BCA	85
Meera Kulkarni	BCA	82
Rahul Sharma	BSc IT	78
Karan Mehta	BCA	74
Rohan Desai	BSc CS	69
Aman Verma	BSc CS	67
Arjun Nair	BSc IT	59

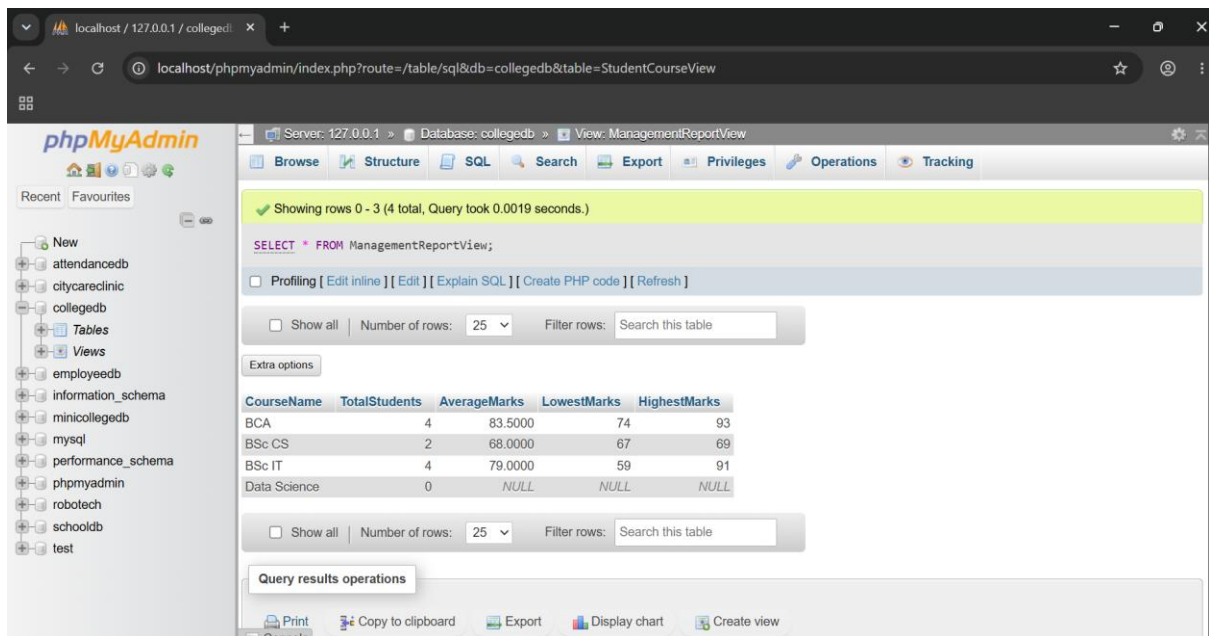
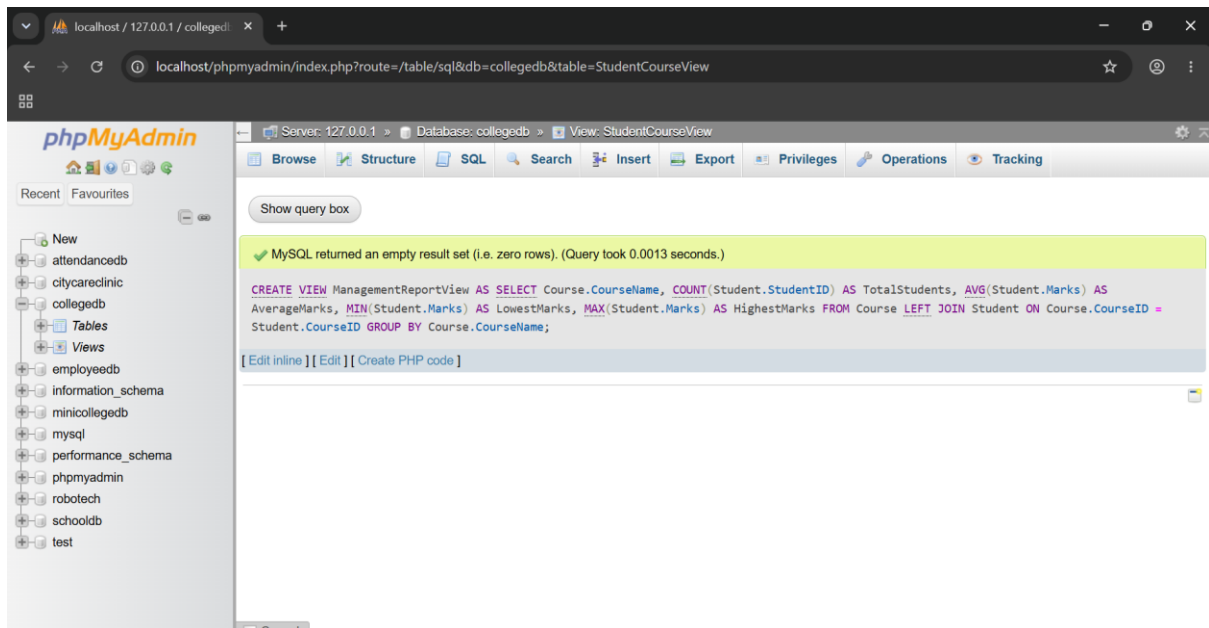
Q6) Create a report view named ManagementReportView.

The view should display course-wise summary:

Column
CourseName
TotalStudents
AverageMarks
LowestMarks
HighestMarks

Also write one line explaining why such a view is useful in reporting.

Answer:-



One-line Explanation

ManagementReportView is useful because it provides a simple course-wise summary report without repeatedly writing the complex SQL query.